Scalability & Future Plan

SMARTBRIDGE - Scalability & Future Plan

- Date: 21/09/2026
- Team ID: SWTID-2026-4223
- Team Size: 4
- Team Leader: Sweatha P
- Team Members:Lakshmi Narayanan R, Rithika P, Sandhiya K
- Project Name: AI FitTrack
- **Maximum Marks:** 1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Local Server Runtime	API is hosted on local Node.js engine (localhost:5000), restricting public internet accessibility.	High
2	External AI Rate Limits	Heavy concurrent API calls to Gemini AI may hit external API quota limits.	Medium
3	No Visual Frontend UI	Project currently operates as a REST API consumed via Postman / Thunder Client.	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution

1	User Load	Single-instance Express Node.js server.	Deploy to Render / AWS ECS with PM2 cluster mode and auto-scaling.
2	Data Storage	MongoDB Atlas M0 Free Tier Cluster.	Upgrade to MongoDB Atlas Dedicated M10 Cluster with indexing on user and workoutDate.
3	Performance	Direct call to Gemini AI for every recommendation.	Implement Redis caching layer for repeated workout plan queries to reduce latency.
4	Security	Standard JWT Auth & Bcrypt Hashing.	Implement express-rate-limit, CORS origin restriction, and Refresh Token rotation.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment & CI/CD	Q4 2026	Deploy backend API to Render/AWS with automated GitHub Actions testing pipeline.
Phase 2	Frontend Web Dashboard	Q1 2027	Build React.js frontend interface with charts for workout statistics and AI recommendations.

Phase 3	Wearable Device Integration	Q2 2027	Sync realtime step and heart rate data from Apple HealthKit / Google Fit APIs.
Phase 4	Multimodal Vision AI	Q3 2027	Use Gemini Vision API to analyze food images and auto-calculate meal calories.